

Truss Bridge Builder

Demo 8 of 16 · Part four, Example projects · Engineering · 30 minutes for stages 1–3, plus the Sandbox

You need: `index.html` open in any modern browser — projector, Chromebook or phone, mouse or touch. No network, no install, no accounts; nothing is fetched at run time and nothing leaves the machine.

Learning goals

One per stage, in the words the learner reads on the Takeaway card. Each is what its stage demonstrates, and each stage's check-yourself item tests exactly that. The game underneath all four is the same: **carry the load for less than par**.

- **Mechanisms.** A frame with a four-sided panel folds without straining any member; a triangle cannot change shape without changing a member's length — and the solver finds the fold as a rank deficiency, not as a rule someone typed in.
- **Compression buckles.** Compression capacity falls as $1/L^2$ — 180 kN at 2 m, 80 kN at 3 m, 45 kN at 4 m — while tension stays at 300 kN at any length, which is why cheap designs shorten their struts or hang the deck so the long members are pulled.
- **Load paths.** Load splits by stiffness, not by counting: a stiffer path attracts more force, a mid-span pier turns one 16 m span into two 8 m ones and cuts the chord forces far more than in half, and a member drawn grey is steel you paid for that carries nothing.
- **Sandbox.** Every result on this span comes from the same three facts: triangles hold shape, compression capacity falls with L^2 , and load follows stiffness.

Where this sits on the path

The second of Part four's AI-built classroom tools and the second of its games, after Orbital Mechanics Golf. Demo 7 hands forward one habit — commit a prediction before you run anything — and this demo makes it structural: before every load test the learner calls which member goes first, and the engine's own record answers. The solver is the real direct stiffness method, so every collapse is a computed result.

One question for the session

Ask it before anyone touches a control; return to it at the debrief.

"When a bridge fails, which piece goes first — and could you have called it before the truck rolled on?"

Most of the room expects the thinnest-looking member to snap. The demo's answer is that the frame usually folds before anything snaps, and when something does go it is a long *pushed* member, not a pulled one. Both surprises are the statics.

Running it

Open ⚙ **Settings** → **Presentation mode** first: it scales the page for the back row and adds a **Notes** button that reopens this guide. The **Guide** card opens on load; dismiss it with **Start with Mechanisms**. Reset (in Settings) puts the demo back to Level 1 in place — no reload — and **leaves the class leaderboard alone**; the two-step clears under *Class data* are the only way to empty that.

- **0–4 Stage 1, Level 1.** Build *only* the flat roadway. Take the room's call under the canvas — most say "it holds" — and press ► **Load test**. It folds. Name what they saw: joint, member, tension, compression, support, mechanism. Add a post and two struts; run it again, and read the cost against par.

- **4–10** Pairs play Level 1 with **X-ray** on. Have them use **Check a joint**: every force meeting there and their sum, always zero. That panel *is* the method of joints.
- **10–18** **Level 2 — the lesson.** It opens with a rectangular frame already built. The *Notice* line under the canvas says the stiffness matrix is singular; the dashed ghost rocking on the canvas is the fold the solver found. Call it, test it, watch it fold into a parallelogram. Let them add diagonals and watch the cue flip to *Rigid*. Par forces a second pass: which members do nothing?
- **18–25** **Stage 2, Level 3 — Heavy Haul.** Four times the load. Point them at **Weakest link** under the canvas and the debrief's **buckles at** column: compression rating depends on length and tension rating does not.
- **25–30** **Stage 3.** Project a student's debrief. Load Pratt, Howe and Warren from the **Gallery** at equal depth and compare where the forces go. Post the leaderboard. Levels 4–5 and the Sandbox make good homework — the gate opens for free play once a stage has been tested once.

Stage 1 · Mechanisms (Levels 1–2)

Question. Why does a flat roadway fold under a car before any member is overloaded?

Predict prompt. Lay the flat roadway alone and call it: does it hold, snap, buckle, or fold?

Activity. ~2 min Draw the deck; call; test. **~4 min** Level 2's frame: call; test; add diagonals until the cue reads *Rigid*; test again.

Expected observation. The flat deck and the rectangle both fold with every member at 0% of its limit. The debrief's *How it moved* table names the joints that ran away. One diagonal per four-sided panel removes exactly one fold; the cue counts them down.

Check-yourself key. *The fold changed no member's length, and a member only resists a change in its own length.* "Too weak" and "loose joints" are both refuted: strength never entered it, and the pins are ideal.

Teaching limit. Every joint is an ideal pin. Real connections carry some bending and can fail; glued or nailed models usually fail at the joint, which this solver cannot show.

Stage 2 · Compression buckles (Level 3)

Question. The same 4 m member holds 300 kN pulled. Why does it fail at 45 kN pushed?

Predict prompt. A 125 kN truck over 10 m: call the first failure of your first design.

Activity. ~3 min Build or load a Gallery truss; turn on **Numbers**; call; test. **~4 min** Flip a working design under the deck and test again.

$$P_{cr} = \pi^2 EI / L^2 \approx 720 \text{ kN}\cdot\text{m}^2 / L^2, \text{ capped at 180 kN}$$

Length	1 m	2 m	3 m	4 m
Pulled ($\sigma_y A$)	300 kN	300	300	300
Pushed (Euler, capped)	180 kN	180	80	45

Expected observation. The first member to go is a long blue one, and the weakest-link line reports it buckling at a fraction of the 300 kN the same member would hold pulled. The regression suite measures the propped design peaking at 37% of limit and the same steel hung at 32%.

Check-yourself key. *No — a pushed member fails by buckling sideways, at a load that falls with the square of its length.* Second item: *the long members are now in tension, so the 300 kN limit applies.*

Teaching limit. Euler with a 0.6 cap is a stand-in for real column curves; effective length is taken as the whole unbraced run with $K = 1$; the section values are the authors' choices (Details labels which are sourced).

Stage 3 · Load paths (Levels 4–5)

Question. How does a wheel's weight find its way to the ground when you cannot build below the deck, or when a pier appears mid-span?

Predict prompt. Level 4 forbids building below the deck: call the first failure of your first design there.

Activity. ~4 min Gallery bowstring or Pratt at 2–3 m depth; **Check a joint** on a deck joint; call; test. ~3 min Level 5: build two short bridges that share a deck.

Expected observation. In a Pratt the verticals are blue and the diagonals red; in a Howe the reverse; in a Warren the diagonals alternate. On Level 5 the pier drops par from \$3,600-class spans to the forces of two 8 m ones. Grey members are the debrief's under-5% rows.

Check-yourself key. *Not necessarily — a stiffer added path attracts more force and can overload its neighbour. Second item: the verticals carry each panel's shear to the top chord; erase one and its panel becomes a rectangle.*

Teaching limit. The moving load is quasi-static — no dynamics, no impact — and the deck members carry wheels straight to their two joints with no bending.

Stage 4 · Sandbox (Level 6)

Question. With no par and any vehicle, which of your ideas about trusses survive a 20 t crane?

Predict prompt. Pick a Gallery truss and a vehicle; call it before the test.

Activity. Free build on 18 m; vary **Depth** and **Panels**; swap vehicles. Nothing is saved to the leaderboard.

Expected observation. Deeper trusses lower the chord forces; more panels shorten the struts; the crane finds the long compression members first.

Check-yourself key. *Shortening the compression members, or hanging the deck so they pull instead of push. "Shallower" is refuted: chord force is roughly bending over depth.*

Teaching limit. The vehicles are invented, sized to make par interesting.

Misconceptions

"More members always makes it stronger." Load shares by stiffness, so an added member can pull force onto a neighbour and overload it, and every member costs steel and a joint. The debrief flags members under 5% of limit. *Try it:* add one to a design that just made par and watch the peak move.

"The verticals do nothing — they're spacers." In a Pratt they carry each panel's shear in compression; in a Howe, in tension. A member genuinely carrying nothing is drawn grey, and that is a different thing. *Try it:* erase one vertical with X-ray on.

"Tension and compression are just plus and minus." The same 4 m member holds 300 kN pulled and 45 kN pushed. The sign selects a different failure mode. *Try it:* rebuild a design upside down under the deck.

"It failed because the steel was too weak." On Levels 1 and 2 the frame folds with every member at 0% of its limit. Strength cannot fix a mechanism; a diagonal can.

Why connections are priced

Costing follows a fabricator's bill: **steel by weight** (9.4 kg/m at \$4.80/kg, about \$45 a metre) plus **\$180 per joint** for the gusset, bolts and labour. A flat price per metre makes extra panels free, which hides a real design driver; here connections run roughly 45% of a typical bill, so bracing a panel you already have costs steel only while adding a panel costs a joint too. The figures are illustrative; the shape of the bill is the point.

Static determinacy, plainly

Count members m , reactions r , joints j ; each joint gives two equations. $m + r < 2j$: a mechanism — it folds and strength cannot help. $m + r = 2j$: determinate — every member is critical. $m + r > 2j$: indeterminate — load now splits by stiffness, which is why the game solves $K \cdot u = F$ instead of walking joint to joint. The count is necessary, not sufficient: members in the wrong places leave a local mechanism, and the solver checks the matrix, not the count.

Discussion questions

1. The bare rectangle folded without any member changing length. Where did the load go, and why could no member resist it?
2. Two designs, same cost and same joints: one props the deck from above, one hangs it from below. Predict which is safer, then check. What in $P_{cr} = \pi^2 EI / L^2$ explains it?
3. A Warren's diagonals alternate tension–compression along its length. Take one bottom joint and use $\Sigma F_y = 0$ to show why its two diagonals must have opposite signs.
4. Why is the bottom chord's force greatest at mid-span and near zero at the supports?
5. One member snapped and the whole bridge came down. What does that say about its spare load paths, and what would a spare path cost?
6. Turn on **Dead load** and retest a design that just made par. What happens, and why does it matter more as spans grow?

Levels, par and logistics

#	Level	Span / vehicle	Par	Stage
1	First Crossing	6 m · car 30 kN	\$1,050	1 Mechanisms
2	The Long Gap	12 m · car 30 kN	\$2,800	1 Mechanisms
3	Heavy Haul	10 m · truck 125 kN	\$2,800	2 Compression buckles
4	High Water	12 m · van 60 kN	\$3,100	3 Load paths
5	Island Pier	16 m · truck 125 kN	\$3,600	3 Load paths
6	Sandbox	18 m · your choice	—	4 Sandbox

Every par is beatable by a triangulated design and by no un-triangulated ladder at any depth or panel count — verified by `test-physics.js`, 86 checks, deterministic.

Presenter controls

- ⚙ **Settings**: **Open Presenter Notes** (this guide), **Presentation mode**, **Reset** (Level 1, in place, leaderboard kept), then Show par budgets, Keyboard editing, pick-the-vehicle demos, and the two-step leaderboard clears.
- **Details** in each stage intro: what is live, how it works, assumptions, sources, boundary.
- Keys: **1 2 3** Draw / Erase / Check a joint · **Ctrl+Z** undo · **Space** test (once a call is committed) · **X** X-ray · **F** numbers.
- Boards and saved designs live in each machine's browser storage.

Truss Bridge Builder (Bridge Works) · Bryant Harrison · Murray State University · one HTML file, runs offline. Physics: 2D pin-jointed truss by the direct stiffness method; mechanisms detected as rank deficiency and animated along the null-space mode; Euler buckling governs compression; overloaded members are removed and the structure re-solved. Sources for every cited figure are in the app's Details drawer with the date read.